

ST JOHNS WEST, NL, Canada

WMO: 712500

Lat: 47.5133N

Lon: 52.7833W

Elev: 110

StdP: 100.01

Time Zone: -3.50 (NAF)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.3	-11.5	-17.8	0.8	-11.3	-15.9	0.9	-9.8	14.3	-1.7	12.9	-1.2	4.2	290	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.0	26.0	20.0	24.5	19.1	23.0	18.3	21.7	24.2	20.6	22.9	19.6	21.8	5.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.6	23.1	19.8	14.7	22.1	18.7	13.7	21.0	63.7	24.3	60.0	23.0	56.5	21.8	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	10.0	8.9	DB	-16.2	28.8	2.0	1.7	-17.6	30.0	-18.8	31.1	-19.9	32.0	-21.4	33.3	(4)
(5)			WB	-16.6	23.2	2.0	1.3	-18.0	24.1	-19.1	24.8	-20.2	25.5	-21.6	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-2.9	-3.5	-1.7	2.6	6.8	11.3	17.0	17.1	13.4	8.4	4.2	-0.4		
	DBStd	8.13	4.19	4.15	3.76	3.50	3.65	4.14	3.65	3.03	3.53	3.78	3.89	4.22		
	HDD10.0	2099	399	379	363	222	114	32	2	0	8	78	179	322		
	HDD18.3	4523	657	612	622	471	357	213	68	58	151	309	424	580		
	CDD10.0	674	0	0	0	2	16	72	220	222	111	28	4	1		
	CDD18.3	56	0	0	0	0	0	3	28	21	4	0	0	0		
	CDH23.3	262	0	0	0	0	1	18	143	82	18	0	0	0		
	CDH26.7	24	0	0	0	0	0	1	15	6	3	0	0	0		
	WSAvg	4.3	4.8	5.1	5.0	4.5	4.1	3.8	3.7	3.5	3.8	4.1	4.4	4.8		
	PrecAvg	1680	193	171	159	132	112	96	79	115	119	155	159	173		
(15) Precipitation	PrecMax	2000	304	333	319	238	204	183	149	308	218	279	280	305		
	PrecMin	1394	106	58	75	26	24	22	20	39	26	70	86	93		
	PrecStd	169	57	67	56	60	48	39	35	64	47	57	52	60		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	9.6	10.5	16.6	21.2	25.3	28.4	27.3	25.5	20.5	16.9	13.4		
		MCWB	11.3	8.9	8.4	12.0	15.2	19.3	21.5	20.1	20.0	18.4	15.6	12.4		
	2%	DB	8.0	6.6	7.6	13.3	18.4	23.2	26.6	25.4	22.7	18.2	14.8	10.6		
		MCWB	7.5	6.2	5.8	10.2	13.6	17.6	20.4	19.7	18.9	16.4	14.0	10.0		
	5%	DB	4.8	4.1	5.0	10.6	16.0	21.0	24.9	24.0	20.9	16.4	12.8	7.7		
		MCWB	4.3	3.4	3.6	8.0	11.7	16.0	19.6	19.0	17.8	14.8	12.0	7.1		
	10%	DB	2.6	2.2	3.1	8.2	13.5	18.7	23.3	22.6	19.3	14.5	10.6	5.3		
		MCWB	2.1	1.4	1.9	6.1	9.9	15.0	18.3	18.3	16.6	12.8	9.7	4.7		
	0.4%	WB	11.3	9.1	9.1	14.2	17.3	21.2	23.5	22.6	21.6	19.0	16.0	13.0		
		MCDB	11.9	9.5	10.2	15.5	19.6	24.1	27.0	25.5	23.6	20.1	16.6	13.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	7.4	6.3	6.2	10.8	14.9	19.4	22.1	21.3	19.9	17.0	14.0	10.1		
		MCDB	7.8	6.6	7.3	12.4	17.4	21.6	24.9	23.5	21.8	17.7	14.7	10.6		
	5%	WB	4.5	3.6	3.9	8.4	12.4	17.3	21.0	20.2	18.5	14.9	12.1	7.2		
		MCDB	4.8	4.0	4.7	10.0	14.6	19.7	23.4	22.4	20.4	16.0	12.7	7.7		
	10%	WB	2.2	1.5	2.1	6.3	10.4	15.3	19.8	19.2	17.0	13.1	9.8	4.7		
		MCDB	2.6	2.1	2.9	8.0	12.9	18.2	22.1	21.6	18.8	14.3	10.4	5.2		
	5% DB	MDBR	6.7	6.9	6.6	7.3	8.6	8.9	9.2	8.0	8.4	6.9	6.5	6.0		
		MCDBR	10.0	9.5	9.2	11.3	12.9	12.4	10.9	10.3	9.9	8.9	9.5	9.3		
		MCWBR	10.2	9.6	8.1	8.6	8.5	7.6	6.2	6.1	6.6	7.4	9.2	9.3		
		MCWBR	10.1	9.4	8.9	10.4	11.2	10.8	9.3	8.9	9.1	8.5	9.4	9.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.4	9.7	8.4	8.7	8.6	7.6	6.2	6.4	7.0	7.7	9.2	9.5		
	Clear-Sky Solar Irradiance	taub	0.316	0.318	0.336	0.367	0.371	0.403	0.433	0.399	0.371	0.346	0.325	0.313		
		taud	2.354	2.423	2.413	2.365	2.414	2.347	2.303	2.413	2.461	2.487	2.457	2.371		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	754	848	886	886	893	862	828	845	838	804	746	709		
		Edh at Noon	72	83	98	113	112	121	125	107	93	78	66	64		
(44) All-Sky Solar Radiation	RadStd		0.94	1.72	2.89	3.93	4.82	5.08	5.11	4.32	3.44	2.09	1.18	0.79		
			0.07	0.13	0.22	0.39	0.39	0.40	0.37	0.28	0.21	0.19	0.16	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page